Linear Algebra - Homework 1

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Section A

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Attempted Solution:

251-3207

a) System of Three Linear Equations :-

CPU: $2x_1 + 4x_2 + 6x_3 = 40$

Memory: $x_1 + 3x_2 + 4x_3 = 25$

Storage: $x_1 + 2x_2 + 5x_3 = 30$

b) Matrix Form of the System of Equations :-

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 40 \\ 25 \\ 30 \end{bmatrix}$$

$$A \cdot x = b$$

c) Augmented Matrix :-

$$\left[\begin{array}{ccc;c} 2 & 4 & 6 & 40 \\ 1 & 3 & 4 & 25 \\ 1 & 2 & 5 & 30 \end{array}\right]$$

d) Conversion to Row Echelon Form :-

$R_1' = \frac{1}{2} R_1$ $\qquad$ $R_2' = R_2 - R_1$; $R_3' = R_3 - R_1$

$$\begin{bmatrix} 1 & 2 & 3 & ; & 20 \\ 1 & 3 & 4 & ; & 25 \\ 1 & 2 & 5 & ; & 30 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 3 & ; & 20 \\ 0 & 1 & 1 & ; & 5 \\ 0 & 0 & 2 & ; & 10 \end{bmatrix}$$

$R_3' = \frac{1}{2} R_3$

$$\begin{bmatrix} 1 & 2 & 3 & ; & 20 \\ 0 & 1 & 1 & ; & 5 \\ 0 & 0 & 1 & ; & 5 \end{bmatrix}$$

$\therefore$ Hence, REF is found.

e) Values of Variables:-

Let Type A VM $= x_1$

Type B VM $= x_2$

& Type C VM $= x_3$; from REF, we found that

$\boxed{x_3 = 5.}$

Put $x_3 = 5$ in R_2

$0 + (x_2) + (x_3) = 5$

$\cancel{5} + x_2 = \cancel{5}$

$\boxed{x_2 = 0}$

Put $x_2 = 0$ & $x_3 = 5$ in R_1

$x_1 + 2(0) + 3(5) = 20$

$x_1 = 20 - 15$

$\boxed{x_1 = 5}$

$\therefore$ Hence values of VMs are:-

Type A $= 5$

Type B $= 0$

Type C $= 5$

251-3207

2

f) Verification:

$2x_1 + 4x_2 + 6x_3 = 40$ ... (i)

$x_1 + 3x_2 + 4x_3 = 25$ ... (ii)

$x_1 + 2x_2 + 5x_3 = 30$ ... (iii)

Put values of x_1, x_2 & x_3 in each eq one by one

- $2(5) + 0 + 6(5) = 40$

  $10 + 30 = 40$

  $40 = 40$ eq (i) verified

- $(5) + 0 + 4(5) = 25$

  $5 + 20 = 25$

  $25 = 25$ eq (ii) verified

- $(5) + 0 + 5(5) = 30$

  $5 + 25 = 30$

  $30 = 30$ eq (iii) verified

Type A VMs	5
Type B VMs	0
Type C VMs	5

Python Verification: